DEVELOPMENT PLAN

Hybrid Baybayin Character and Stroke Validation System

Prepared to guide Chapter 3 methodology and system implementation

Core principle

The validator makes its decisions through defined character and stroke checks. The knowledge graph and RAG layer retrieve KWF-grounded evidence and explanations; they do not replace the measurable validation logic.

Research alignment

The plan directly supports the study objectives: (1) build a labeled Baybayin dataset with stroke-level information, (2) develop a stroke-based validation model, and (3) evaluate it against expert-reviewed ground truth.

1. Development architecture

The system is developed in two connected versions. Version 1 establishes reliable character-form validation and the data-capture workflow. Version 2 adds the stroke-level validation required by the thesis.

Component	Purpose	Primary output
KWF reference standard	Defines valid character forms, strokes, and kudlit rules.	Character specification and validation rubric.
Knowledge graph + RAG	Retrieves verified text, images, rules, and later instructional media.	Evidence-based explanation.
Version 1: form validator	Checks whether a completed drawing resembles its intended character.	Form-valid/form-invalid result.
Version 2: stroke validator	Checks count, order, direction, trajectory, and kudlit placement.	Specific formation-error feedback.
Expert evaluation	Reviews labels and validates model output against ground truth.	Performance evidence for Chapter 4.

Implementation flow

KWF rules + reference images -> knowledge graph -> drawing canvas captures final form and online stroke data -> form and stroke checks -> RAG retrieves the relevant rule -> user receives a valid/invalid result with a specific explanation.

2. Iterative development plan

Iteration 1 - Requirements and standards

Activities: Select supported characters; convert KWF material into character specifications; define valid and invalid writing; prepare an expert validation rubric.
Output: Approved character specification and rubric.

Iteration 2 - Knowledge base

Activities: Create a multimodal knowledge graph with character records, reference images, writing rules, kudlit rules, common errors, and sources.
Output: Queryable, source-traceable knowledge base.

Iteration 3 - Version 1 form validator

Activities: Build the drawing canvas; store the final image and raw event data; normalize the drawing; compare the final form with its reference.
Output: Working alphabet/form-conformance prototype.

Iteration 4 - Dataset collection and labels

Activities: Collect correct and controlled-error samples; capture coordinates, timestamps, and pen-down/pen-up events; have an expert review labels.
Output: Expert-reviewed stroke-level dataset.

Iteration 5 - Version 2 stroke validator

Activities: Segment strokes; extract features; compare order, direction, trajectories, spatial relations, and kudlit placement with the reference.
Output: Hybrid stroke-based validator.

Iteration 6 - Evaluation and integration

Activities: Test on held-out participant data; calculate metrics; review errors; connect feedback to the knowledge graph and RAG layer.
Output: Final prototype and Chapter 4-ready results.

3. Chapter 3 methodology mapping

Chapter 3 should document the plan as a reproducible method. It must describe what will be done, the input to each stage, the activity performed, and the output produced - not the final results.

Chapter 3 section	What it documents
3.1 Research Design	R&D; design with an experimental performance-evaluation component; two connected validator versions.
3.2 Participants and Data Sources	Volunteer writers, expert reviewer, and KWF documentary sources.
3.3 Research Instruments	Drawing canvas, task sheet, expert checklist, knowledge graph, and evaluation tools.
3.4 System Development Method	Agile iterations and their testable deliverables.
3.5 Hybrid Model Development	Data acquisition, cleaning, feature extraction, model development, evaluation, and utilization.
3.6 Ethics and Data Management	Informed consent, anonymization, restricted access, and data traceability.

4. Model-development stages

Data acquisition

Inputs: KWF reference material, participant drawings, expert rubric. Activities: build character specifications, capture final images and online trajectories, and review labels. Output: a labeled multimodal dataset.

Preprocessing

Inputs: raw drawings and event records. Activities: remove incomplete records, normalize images and coordinates, segment strokes, preserve relative kudlit position, and resample trajectories. Output: clean and comparable samples.

Feature extraction and retrieval

Inputs: normalized data and knowledge graph. Activities: derive form features, stroke count/order/direction/shape/spatial features, and retrieve supporting KWF references. Output: validation features plus evidence.

Modeling and evaluation

Inputs: features, rules, reference trajectories, and labels. Activities: form comparison; rule checks; trajectory similarity; participant-level data splitting; expert-ground-truth comparison. Output: measured model performance.

Model utilization

Inputs: a user drawing and selected target character. Activities: validate, retrieve relevant rules, and present feedback. Output: a user-facing valid/invalid result with a specific explanation.

5. Evaluation plan

Evaluate the form validator and stroke validator separately, then assess the complete hybrid system. Compare every decision with expert-reviewed ground truth. Report accuracy, precision, recall, F1-score, confusion matrices, and error-specific results for form, count, order, direction, and kudlit placement. Evaluate RAG retrieval separately for source correctness and feedback relevance. Record response time if practical feedback is a project goal.

Immediate next action

Complete Iteration 1 first: choose the initial character set and produce the character specification and expert validation rubric. These define what counts as valid before any model, knowledge graph, or dataset can be trusted.

Implementation decisions to confirm

1. Initial character scope: identify the exact KWF character set and whether kudlit variants are included in the first evaluation.
2. Ground truth: name the qualified Baybayin expert or reviewers and agree on the labeling procedure.
3. Dataset protocol: set the number of writers, repetitions per character, and controlled error categories before collection.
4. Validation thresholds: establish the rule tolerances and trajectory-similarity thresholds using the validation subset, not the final test data.
5. Prototype boundary: state that Version 1 establishes form conformance and Version 2 delivers the thesis stroke-validation claim.